

11/7/2026

Introduction to cryptography

- Cryptography is the art and science of securing digital information through mathematical techniques.
- It converts (plain text) readable data into an unreadable form (cipher text).

Security layers :-

→ Computer Security

Protects computers from cyber attacks.

→ Network Security.

Secures communication between connected devices.

→ Internet Security

Protects online activities such as web browsing, email, cloud services.

→ Cyber Security

Secure computers and electronic devices and systems from cyber attacks.

Basic Concepts :-

Plaintext :- The original intelligible message.

Cipher text :- The transformed message.

Cipher :- An algorithm for transforming by

transposition and substitution methods.

Key :- Information used by the cipher known only to the sender and receiver.

Encoder (encode)

Decoder (decode)

Cryptanalysis :- The principles and methods of breaking without knowledge of the key

Cryptography

Classified into three independent dimensions :-

(i) Type of operations for transforming plain text into cipher text.

- Substitution $\rightarrow$ Plaintext into element.
- Transposition $\rightarrow$ element in plaintext rearranged.

(ii) The number of keys.

- Symmetric key or single key encryption
- Public key encryption.

(iii) The way in which the plain text is post processed

- Block cipher : Input and block of element at a time, output block for each input block.

A stream cipher : Input element continuously.

Aspects of Security:

Security attack

(CIA - Triad) $\rightarrow$ Confidentiality, Integrity, Availability.

Four categories of attack

→ Interruption

Attack on availability

Eg: Destruction of hardware.

→ Interception.

Attack on confidentiality

Eg: Wire tapping.

→ Modification

Attack on integrity

Eg: changing datafile, Altering program.

→ Fabrication.

Attack on authenticity

Eg: Hackers getting access.

(A) Cryptography Attacks

→ Passive attacks

Without altering data.

→ Active attack.

Changing the information.

further classified.

- Masquerade: Pretends to be different entity
- Replay: Passive capture of.
- Modification of message
- Denial of service

(B) Security Services / Goals.

Goal - protective information.

(ii) Authentication $\rightarrow$ verification.

Types:

1. Peer entity Authentication.

Verifies both communicating parties.

(2) Data origin authentication:-

Confirms that data.

3. Integrity

• Ensures data is not modified.

(Types of integrity services:

(a) Connection Oriented Integrity.

• Protects continuous stream of messages.

• detects the denial of service (DOS).

(b) Connectionless oriented integrity.

• Protects individual messages.

• Detects message modifications only.

(c) Access control

Access control limits who can access.

d) Availability

Ensures that services, data and applications are available on demand.

Security mechanism :-

Used to protect data, system, mechanism.

Cryptographic threats.

Cryptographic threat is any potential event or

- a) Brute force attack.
- b) Dictionary attack.
- c) Cryptanalysis attack.
- d) Man in the middle attack (MITM)
- e) Replay attack
- f) side channel attack
- g) Insider Threat.

Cryptographic Risks.

- a) Weak encryption Algorithm.
- b) Weak Key Management.
- c) Weak password
- d) poor random number Generation.
- e) software vulnerabilities.
- f) Human errors
- g) Key loss

Bellare Padua model

Caesar cipher : Shift Key of 3 .

Plain text $\rightarrow$ cipher text

plain text $\rightarrow$ cybersecurity

cypher cyber text : F B E H U V H F X U

Monoalphabetic substitution cipher .

Instead of shifting alphabets by fixed
and random permutation / some fixed is
to the alphabets. alphabets

Key Mapping

A $\rightarrow$ Q, B $\rightarrow$ W, C $\rightarrow$ E, D $\rightarrow$ R, ... X $\rightarrow$ B,

Examples :

Plain text : S E C U R I T Y

Cipher text :

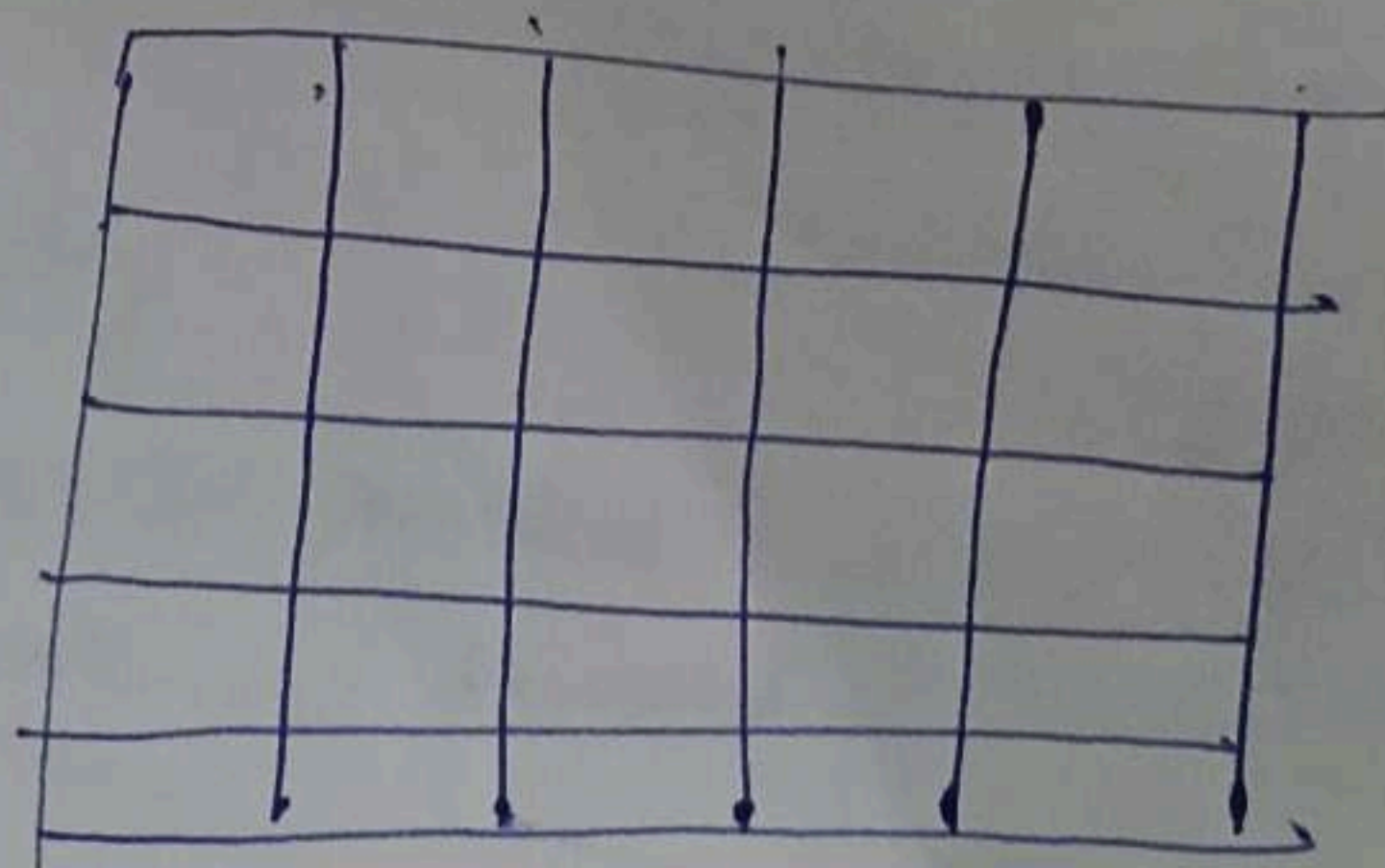
Playfair cipher :

The best known multiple letter encryption
is the playfair .

i . u j are considered as one value ,

they can occupy the same .

Step : 1 For 5 x 5 grid .



step: 2

step 3: if two letters are in the same row, replace with immediate right letters.

* If two letters are the same column, then replace with immediate below ~~letter~~ letters

* If the letters are used in the same row & column, draw a square as corners of P & T the remaining two corners becomes C & T.

eg:

plain text: Good Morning.

key - playfair example:

P	l	a	y	f
i/j	x	e	o	m
b	c	d	g	h
k	n	u	v	s
t	u	v	w	z

go|od|mo|rn|in|
dv|vo|es|cu|rk|a

Plain text: computer.

Key: Security.

S	e	c	u	x
k	t	y	a	b
d	f	g	h	i
l	m	n	o	p

Co|mp|ut|er.

Un|en|ea|sc

CO1 ATI

1. Scenario Event : Communication Security.

A Student organization is preparing to share confidential event details with its members using a simple classical encryption method. During testing a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- Explain how an authorized member, aware of the encoded rule, would retrieve the original message.

2. Scenario : Intercepted suspicious Message.

A cybersecurity intern intercepts the coded message: "GSRH RHZ HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- Deduce the most likely original message.
- Justify your answer by analyzing repeating patterns, letters structures, or known cipher characteristics.

21/7/26

Poly Alphabetic cipher:

- A set of monoalphabetic cipher.
- If a letter is repeated more than once in plain text then that is replaced by the same cipher.
- All other letters are incremented by 1.

Eg: PT: Hello

CT: ifmmp

- In polyalphabetic cipher, the same letter in plain text have different cipher text.

Vigenere cipher:

- Let the sequence of PT, $P = P_0 P_1 P_2 \dots$
- Key, $K = K_0 K_1 K_2 \dots K_{m-1}$, $m < n$
- CT = $C = C_0 C_1 C_2 \dots C_{n-1}$

$$CT = E(K, P)$$

$$= E((K_0 K_1 K_2 \dots K_{m-1}), (P_0 P_1 P_2 \dots))$$

$$= (P_0 + K_0) \bmod 26; (P_1 + K_1) \bmod 26, \dots, (P_{n-1} + K_{n-1}) \bmod 26.$$

- ~~In vigenere cipher.~~

Eg: Key: raju

PT: Welcome to esE depart

$$(r + W) \bmod 26$$

$$(17 + 22) \bmod 26 = (P + K) \% 26$$

$$39 \% 26 = 13 = N$$

Transposition technique

Eg: PT is abc, then CT is bde

Key is some + 1

Railfence:

- Sequence of diagonals of any depths
- Read them as a sequence of rows.

Eg: PT - meet me after the class

Diagonal depth: 2

Row 1	r ₁	m	e	m	a	t	r	h	c	a
Row 2	r ₂	e	t	e	f	e	t	e	t	e

Result CT: mematrhcasetefetels

4 3 1 2 5 6 7 — order

a t t a c ~~R~~ ~~R~~

O S t p O N E

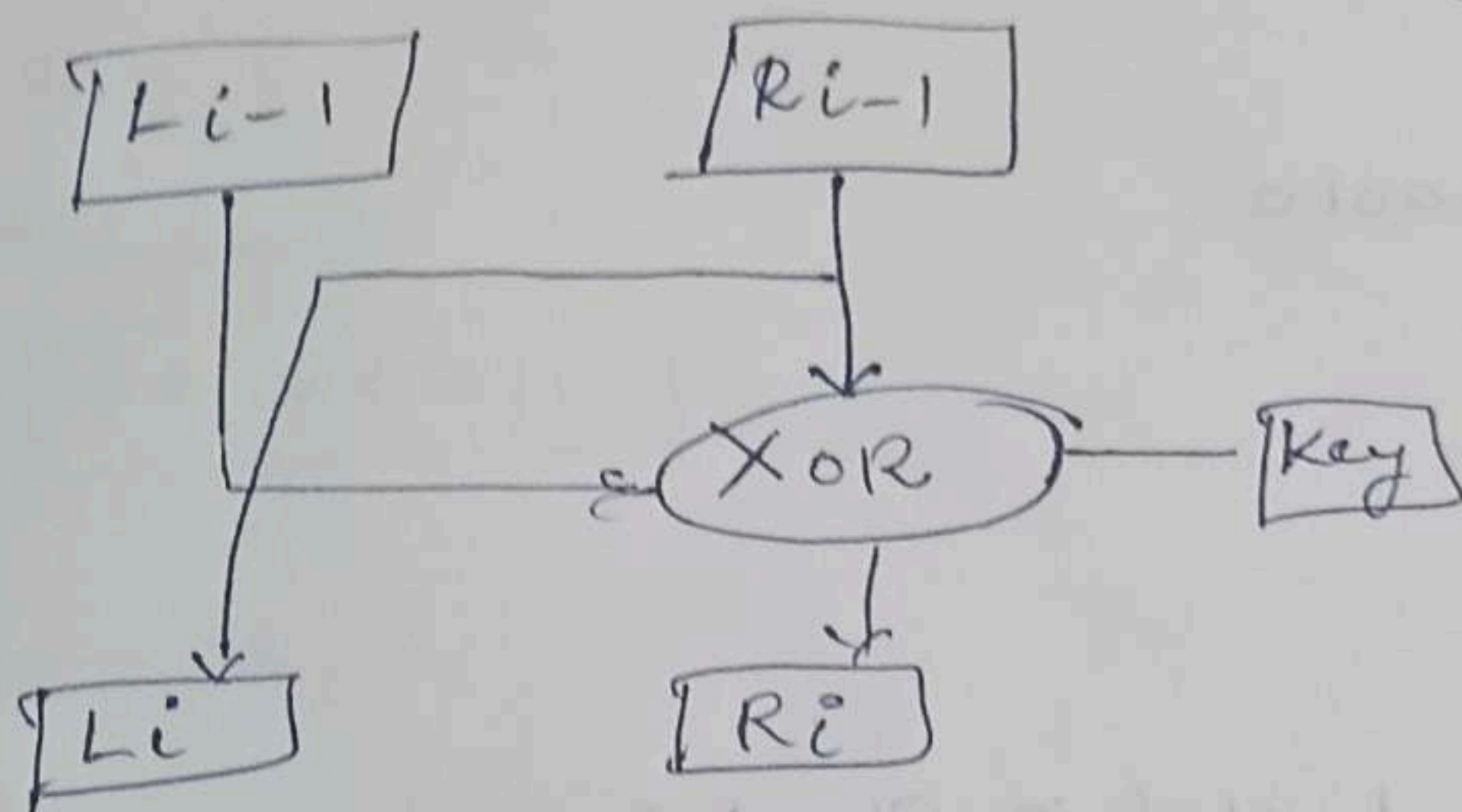
D U N T I L T

W O A M X Y Z

result CT: ttnaa ptm t suoa o dw col
nlypetz.

Feistel cipher structure

$$L_i = R_{i-1}$$



1. Divide the PT into two halves.
2. L_{i-1} , left half and right R_{i-1} , have same structure
3. perform some function that depends on the particular algorithm.
Eg: DES (or) RSA.

Question: 1

A Feistel cipher uses an 8 bit block and one round

plaintext = 10101101

Round Key $K_1 = 1100$

Round function: $F(R, K) = R \oplus K$

Questions:

1. Split the plaintext

Two round Feistel
2. An 8 bit plaintext is encrypted using two rounds.

- plaintext = 11100010.

- $K_1 = 0110$

- $K_2 = 1001$

3. Decryption using feistel structure.

- A ciphertext 00111100 was produced by one feistel cipher.

- Round key $K_1 = 0101$

- $F(R, K) = R \oplus K$.

Questions

1. split the cipher text into L_i and R_i
2. Recover R_0 using the feistel property
3. compute $F(R_0, K_1)$.
4. Recover the original L_0 .
5. Write the recovered plaintext block.

sln. Given:

Property R

Cipher text : 00111100

Round key (K_1) = 0101

Function: $F(R, K) = R \oplus K$

1. $L_1 = 0011$ Split

$R_1 = 1100$.

2. Recover R_0 .

$$R_0 = L_1$$

Therefore

$$\boxed{R_0 = 0011}$$

3. Compute:-

$$F(R_0, K_1)$$

using XOR.

$$\begin{array}{r} 0011 \\ 0101 \\ \hline 0110 \end{array}$$

$$F(R_0, K_1) = 0110$$

4. Recover original L_0 .

$$R_1 = L_0 \oplus F(R_0, K_1)$$

$$L_0 = R_1 \oplus F(R_0, K_1)$$

$$\begin{array}{r} 1100 \\ + 0110 \\ \hline 1010 \end{array}$$

$$L_0 = 1010$$

5. Recovered plain text.

$$\text{Plaintext} = L_0 R_0$$

$$= 10100011$$

R_0	K_1	$R_0 \oplus K_1$
0	0	0
0	1	1
1	0	1
1	1	0

Rough

Encryption General Formula

$$C = E(K, P)$$

$$C = (P + K) \bmod 26$$

Decryption Formula

$$P = (C - K) \bmod 26$$

$$G.F.: P = D(K, C)$$

ASCII

A 65

B 66

C 67

H 72

a 97

b 98

c 99

Hill cipher:

- Hill cipher is a classical polygraphic substitution
- It uses matrix multiplication to encrypt blocks of letters.

Steps

1. Convert letters into numbers
2. Group letters into blocks (usually 2 or 3 letters)
3. Multiply

Hill cipher Algorithm:

1. form 2×2 matrix.
2. form a column vector of length similar to ~~key~~ key matrix.

$$C = Kp \text{ mod } 26$$

$$(C_1 C_2) = \begin{pmatrix} K_1 & K_2 \\ K_3 & K_4 \end{pmatrix} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \text{ mod } 26$$

Letter to Numbers.

A 0

B 1

C 2

D 3

Eg. $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

$$(3 \times 5) - (3 \times 2) = 15 - 6 = 9$$

$\gcd(9, 26)$, matrix is valid.

plaintext.

H I

H = 7

I = 8

$$P = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Decryption.

$$P = K^{-1} \times C \text{ (mod } 26)$$

pbm: 1 Hill cipher.

Plaintext: HELLO.

Letters $\rightarrow$ Numbers.

A 0

B 1

C 2

D 3

.

.

.

H 7

E 4

L 11

L 11

O 14.

X 23.

HELLO $\rightarrow$ 7 4 11 11 14.

Pairing: HE LL O X

$\begin{pmatrix} 7 & 4 \\ 11 & 11 \end{pmatrix}$

$\uparrow$
Padding

Key matrix.

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

HE (7, 4)

LL (11, 11)

OX (14, 23)

Encrypt each pair. For HE.

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \end{bmatrix} = \begin{bmatrix} 21 + 12 \\ 14 + 20 \end{bmatrix} = \begin{bmatrix} 33 \\ 34 \end{bmatrix}$$

Taking modulo.

$$33 \pmod{26} = 7 \rightarrow H$$

$$34 \pmod{26} = 8 \rightarrow I$$

HE $\rightarrow$ HI

for LL:

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 11 \\ 11 \end{bmatrix} = \begin{bmatrix} 66 \\ 77 \end{bmatrix}$$

$$66 \bmod 26 = 14 \rightarrow O$$

$$77 \bmod 26 = 25 \rightarrow Z$$

LL $\rightarrow$ OZ

$$\begin{array}{r} 26 \overline{) 66} \quad 2 \\ \underline{52} \\ 14 \end{array}$$

For OX:

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 14 \\ 23 \end{bmatrix} = \begin{bmatrix} 111 \\ 143 \end{bmatrix}$$

$$42 + 6$$

$$28 + 11 = 1$$

$$111 \bmod 26 = 7 \rightarrow H$$

$$143 \bmod 26 = 13 \rightarrow N$$

OX $\rightarrow$ HN

Final ciphertext:

HELLO.

HE $\rightarrow$ HI

LL $\rightarrow$ OZ

OX $\rightarrow$ HN

Ciphertext = H1OZHN

Decryption:

$$P = K^{-1} \times C \pmod{26}$$

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$(15 - 6) = 9$$

$$\boxed{K=9}$$

$$9 \bmod 26 = 9$$

$$9 \times 3 = 27 \equiv 1 \pmod{26}$$

Adjoint:

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix}$$

convert negative values

$$26 - 3 = 23$$

$$26 - 2 = 24$$

$$\text{so } \begin{bmatrix} 5 & 23 \\ 24 & 3 \end{bmatrix}$$

$$3 \times \begin{bmatrix} 5 & 23 \\ 24 & 3 \end{bmatrix} = \begin{bmatrix} 15 & 69 \\ 72 & 9 \end{bmatrix}$$

$$15 \pmod{26} = 15$$

$$69 \pmod{26} = 17$$

$$72 \pmod{26} = 20$$

$$9 \pmod{26} = 9$$

$$K^{-1} \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

$$P = K^{-1} \times C \pmod{26}$$

$$= \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 15(7) + 17(8) \\ 20(7) + 9(8) \end{bmatrix} = \begin{bmatrix} 241 \\ 212 \end{bmatrix}$$

$$241 \bmod 26 = 7$$

$$212 \bmod 26 = 4$$

$$7 = H$$

$$4 = E$$

$$HI \rightarrow HE$$

For OZ.

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \times \begin{bmatrix} 14 \\ 25 \end{bmatrix} = \begin{bmatrix} 635 \\ 505 \end{bmatrix}$$

$$635 \bmod 26 = 11$$

$$505 \bmod 26 = 11$$

$$11 = L$$

$$OZ \rightarrow LL$$

For HN.

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 13 \end{bmatrix} = \begin{bmatrix} 326 \\ 257 \end{bmatrix}$$

$$326 \bmod 26 = 14$$

$$257 \bmod 26 = 23$$

$$14 = O$$

$$23 = X$$

$$HN \rightarrow OX$$

plaintext = HELLOX.

23/

25/7/26

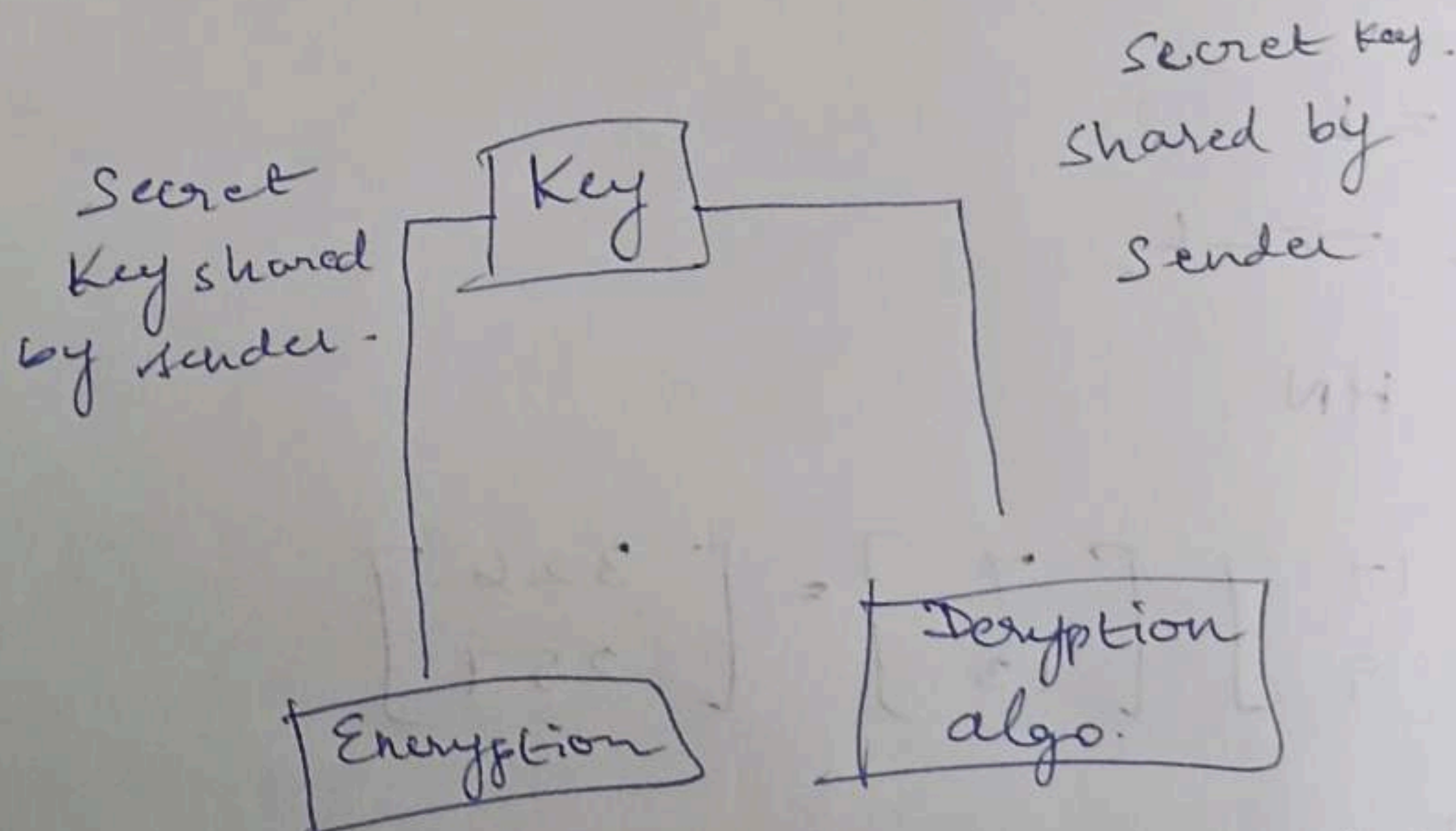
Unit-2 : Conventional Cryptography

Symmetric Key Cryptography :

- Cryptosystem / Secret Key Cryptography - / Private
- Same secret Key for both encryption and decryption.

Techniques used in symmetric Key Cryptography :-

- Substitution techniques.
- Transposition techniques.



Types of Symmetric Key :

1. • Block cipher: Encrypts fixed size blocks.

• Eg: 64 bits and 128 bits.

• DES, 3DES, AES etc.

2. stream cipher:

Encrypts one bit or one byte

eg: RC4

Simplified data encryption Standard :- (SDES)

Encryption

Plain text : 8 bits

Key size : 10 bits

Cipher text : 8 bits

Similarly

Cipher text : 8 bits

Key : 10 bits

Plain text : 8 bits

Example : of Encryption or decryption using
(SDES)

Problem :-

Key : 1010000010

Plaintext : 111000001

Step : 2 :-

Then the key should proceed with permutation

P10 :

Input order	1	2	3	4	5	6	7	8	9
output order	3	5	2	7	4	10	1	9	8

put the key 1010000010 into the P10 table

Output	1	2	3	4	5	6	7	8	9
Input	1	0	1	0	0	0	0	0	1
output	1	0	0	0	0	0	1	1	0
order	3	5	2	7	4	10	1	9	8

so output of the Key after p10 is Key.

output: 1 000001100.

Step: 3

divide the Key into two halves, the left half and right half.

output: 1 000001100.

1st half: 10000

~~2nd half~~

2nd half: 01100.

Step: 4:

round shift: one bit round shift on each half.

Before 1 - 10000.

2 - 01100.

After 1 - 00001

2 - 11000.

After shift combine both.

0000111000

11000

Step: 5.

Order:	1	2	3	4	5	6	7	8	9	10
combined bit	0	0	0	0	1	1	1	0	0	0
output order	6	3	7	4	8	5	10	9		
output	1	0	1	0	0	1	0	0		

K1 = 10100100.

Step - 6.

- We require two keys because S-DES has two rounds.
- Create Second Key.

Copy both of the halves in step 4: i.e. the result of the first round shift.

Skip the unused bits.

1st half: 00001

2nd half: 11000

Step - 7.

Performing two bit round shift.

After 2 bit round shift:

1st = 00100

2nd = 00011

Combine: 0010000011

Step - 8.

order:	1	2	3	4	5	6	7	8	9	10
Input:	0	0	1	0	0	0	0	0	1	1
output order:	6	3	7	4	8	5	10	9	2	
output	0	1	0	0	0	0	1	1		

$K_2 = 01000011$

ii) Encryption of plain text into ciphertext in SDES.

The input & output is 8 bits or.

The block size is always 8 bits.

plain text: 01110010

Step 1:

performing $Ip-8$ initial permutation on plaintext

output bits : 10101001

Order :	1	2	3	4	5	6	7	8
Input Bits :	1	0	1	0	1	0	0	1
Permute bits :	2	6	3	1	4	8	5	7
output :	0	0	1	1	0	1	1	0

Step : 2

Divide into 2 halves -

~~1st - 0011~~

1st = 1010

2nd - 1001

Step : 3. 2nd half

Right half :	1	0	0	1				
Order :	1	2	3	4	5	6	7	8
Expanded bits :	4	1	2	3	2	3	4	1
Output :	1	1	0	0	0	0	1	1

Step 4:

Take the output and XOR with the 1st Key, K_1

Key = 1: 10100100 (Step 5 Key generation)

Output bits : 11000011 (Step 3 PT)

Key 1 XOR bits : (10100100) XOR (11000011)

D/P = 01100111

Step: 5.

Now put each half into the S-boxes, two boxes.

S-0 and

S-0

For left half: 0110.

Row: 00 = 0 (0th row)

col: 11 = 3 (3rd col)

So, 2 = 10

S-1

For right half: 0111

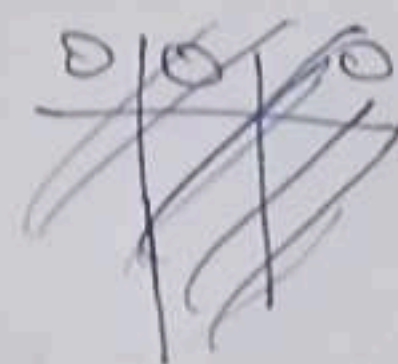
Row: 01: 1 (1st row)

col: 11 = 3 (3rd col) So S1 = 3 = 11.

S-0			
1	0	3	2
3	2	1	0
0	2	1	3
3	1	3	2

S			
0	1	2	3
2	0	1	3
3	0	1	0
2	1	0	2

0	0	0
0	1	1
1	0	2
1	1	3



Step: 6.

Performing permutation: 4.

Now the output is 0111

order					
input					
output order					
output					

Order	1	2	3	4
input	1	0	1	1
output order	2	4	3	1
output	0	1	1	1

Step: 7

~~XOR this out~~

$$\begin{array}{r}
 0111 \\
 \oplus 1010 \\
 \hline
 1101
 \end{array}$$

output: 1101

Step: 8

Now combine the output (1101) with right half of initial permutation (step 2)

Right half permutation of step 2: 1001

8 bit is 11011001

Step: 9

Dividing into 2 halves and swapping them, After

Left half: 1001

Right half: 1001

Step: 10

8 bits = Bits: 10011001